

PROGRAM FLOW DIAGRAM

BOOT

MAIN LOOP

Boot

System Initialization

Wait for steady I2S connection
Mount SD card with automatic retry on failure
Configure audio and stream format (44.1kHz, 32-bit, mono)

Open Session Folder

Create uniquely numbered session directory
All subsequent files stored here for simple data batching

SD Write
Available?

SD Write Cycle

5 Seconds Start-End

- Open new sequentially numbered .wav file
- Route raw PCM I2S data through audio encoding pipeline
- Close file to ensure data safety

Error Handling Delay

5 Seconds Start-End

Pauses full program execution

Sleep

55 Seconds

ESP32 enters a low-power state, Only clock and I2S signal are left active

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